

PRANAV S BHAT

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OBJECTIVE

Backend & AI Systems Engineer experienced in building scalable backend systems, Retrieval-Augmented Generation (RAG) pipelines, and multi-agent AI workflows. Proficient in Python (FastAPI), REST APIs, LLM integration, vector search (FAISS), and workflow orchestration for real-time AI applications. Skilled in backend optimization, concurrent request handling, authentication, and modular system design, with strong foundations in DSA, DBMS, and intelligent automation systems.

EDUCATION

B.E in Information Science | Vidyavardhaka College of Engineering, Mysuru | Expected 2027
CGPA:7.2

Pre university | Sankalpa Gopalaswamy, Mysuru
Percentage: 91.3%

Secondary Education | Maharshi Public School, Mysuru
Percentage:93.6%

SKILLS

Languages | Python, JavaScript, SQL.

System Design & Scalability | Scalable Backend Systems, Concurrent Request Handling, Layered Architecture, Microservices Basics.

API & Backend | REST APIs, JSON APIs, Authentication, Postman, FastAPI.

Frameworks/Libraries | Pandas, Pydantic, LangGraph, Tensorflow, LlamaIndex.

Agentic AI | RAG, LLMs, Embedding Pipelines, Agentic Workflows, Vector-Search.

Concepts | DSA, DBMS, Computer Networks.

Soft Skills | Communication, Leadership, Creative Thinking, Time Management.

PROJECTS

FAISS-RAG Backend for Grounded LLM Responses

- Developed a scalable RAG system using SentenceTransformers and FAISS for semantic search and document retrieval.
- Achieved <200ms retrieval latency through optimized indexing and efficient request handling.
- Integrated Llama 3.2 via Ollama for context-aware response generation, reducing hallucinations and improving response relevance.

Agentsphere

- Architected and developed a multi-agent backend platform using FastAPI and PostgreSQL to orchestrate AI/ML workflows.
- Designed agent orchestration workflows supporting task scheduling, execution tracking, and lifecycle management.
- Built scalable backend services supporting concurrent execution and workflow automation.

TECHNICAL PAPER

- Implemented an agent-based simulation system with dynamic state transitions (healthy, infected, recovered) to model large-scale population behavior and workflow interactions.
- Developed a scalable simulation engine supporting configurable parameters such as infection rate, mobility, and population density for real-time scenario analysis.
- Built real-time data processing and visualization workflows to efficiently manage continuous state updates across large populations.

EXTRACURRICULAR

- Campus Brand Ambassador – Delhivery, leading outreach and engagement initiatives.
- Winner – Infothon National-Level Hackathon.
- Active member of Google Developer Groups (GDG) and U&I Trust.